

WAZUH + VIRUSTOTAL

Automated Malware Detection and Removal on Windows

Practical Lab Report

LAB OBJECTIVE

Detect an EICAR test file with Wazuh FIM, check the file hash using VirusTotal, and automatically remove the detected file using a Windows Active Response executable.

Author: Shushant | Date: 26 September 2026 | Environment: VMware lab

1. Overview

File Integrity Monitoring records file creation, modification, and deletion. In this lab, Wazuh monitored the Windows Downloads folder, sent the detected file hash to VirusTotal, and launched a custom executable to remove a malicious test file.

Detection flow

File added (554) → checksum changed (550) → VirusTotal detection (87105) → Active Response → file deleted (553).

2. Lab Requirements

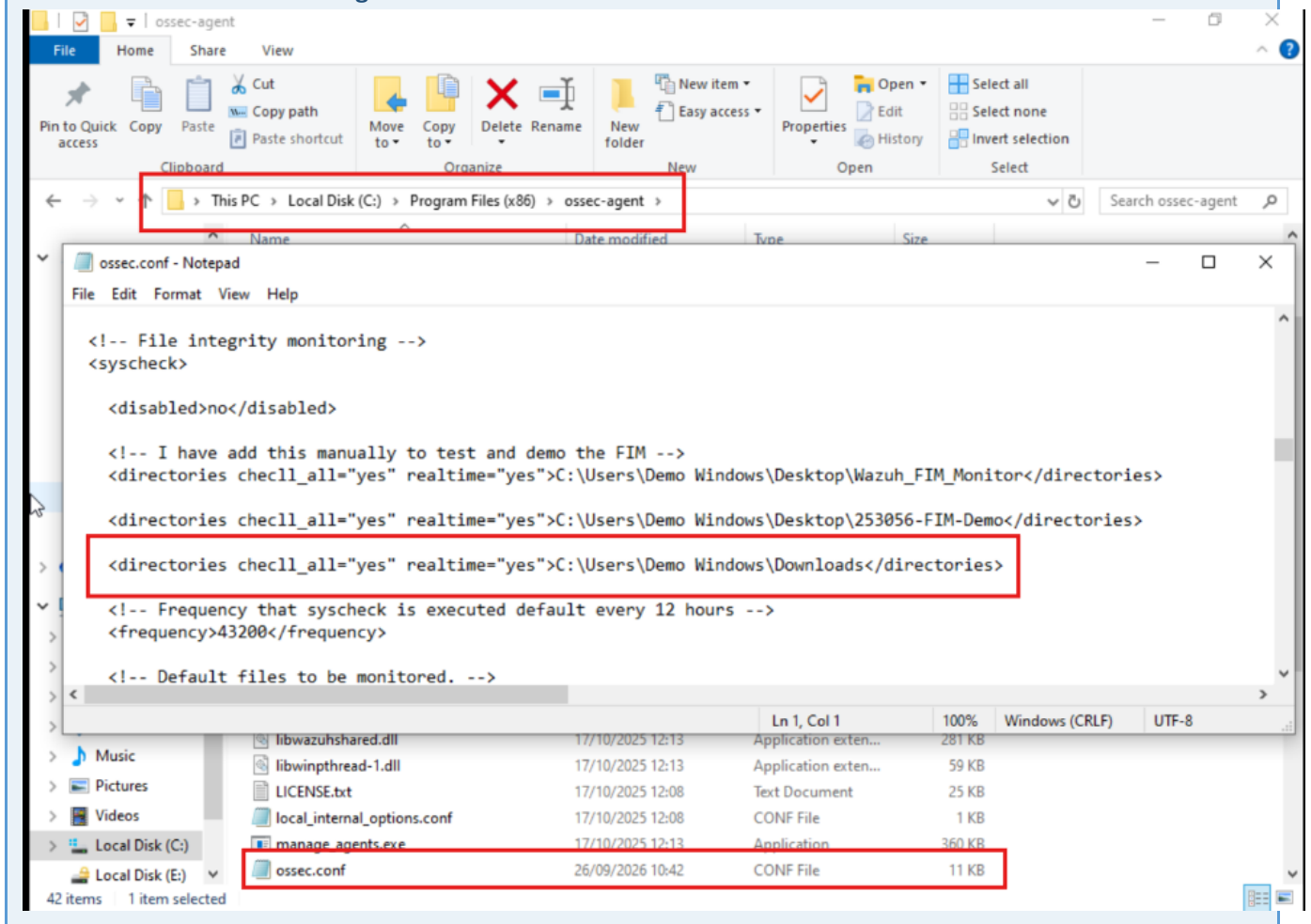
- Ubuntu Wazuh Manager with internet and DNS access.
- Windows Wazuh agent connected to the manager.
- VirusTotal API key (You have sign on [VirusTotal Website](#) which is completely free). The key must be hidden in public screenshots.
 - Limitation on VirusTotal free account:
 - 4 requests per minute
 - 500 requests per day
- Python and PyInstaller are installed on Windows.
 - In windows powershell:
 - *python -m pip install pyinstaller*
- [EICAR](#) anti-malware test file for safe validation.

3. Configure FIM on Windows

The Windows Downloads directory was monitored in real time so newly downloaded files would generate FIM events immediately.

```
<syscheck>
  <disabled>no</disabled>
  <directories check_all="yes" realtime="yes">C:\Users\Demo Windows\Downloads</directories>
</syscheck>
```

FIGURE 1: Windows FIM configuration



Why this step?

Without FIM, Wazuh has no file event or hash to send to VirusTotal. Restart the Windows Wazuh service after editing ossec.conf.

4. Configure VirusTotal and Active Response

The manager queried VirusTotal for syscheck events. Rule 87105 then triggered remove-threat.exe on the Windows endpoint.

```
<integration>
  <name>virustotal</name>
  <api_key>YOUR_API_KEY</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>

<command>
  <name>remove-threat</name>
  <executable>remove-threat.exe</executable>
  <timeout_allowed>no</timeout_allowed>
</command>

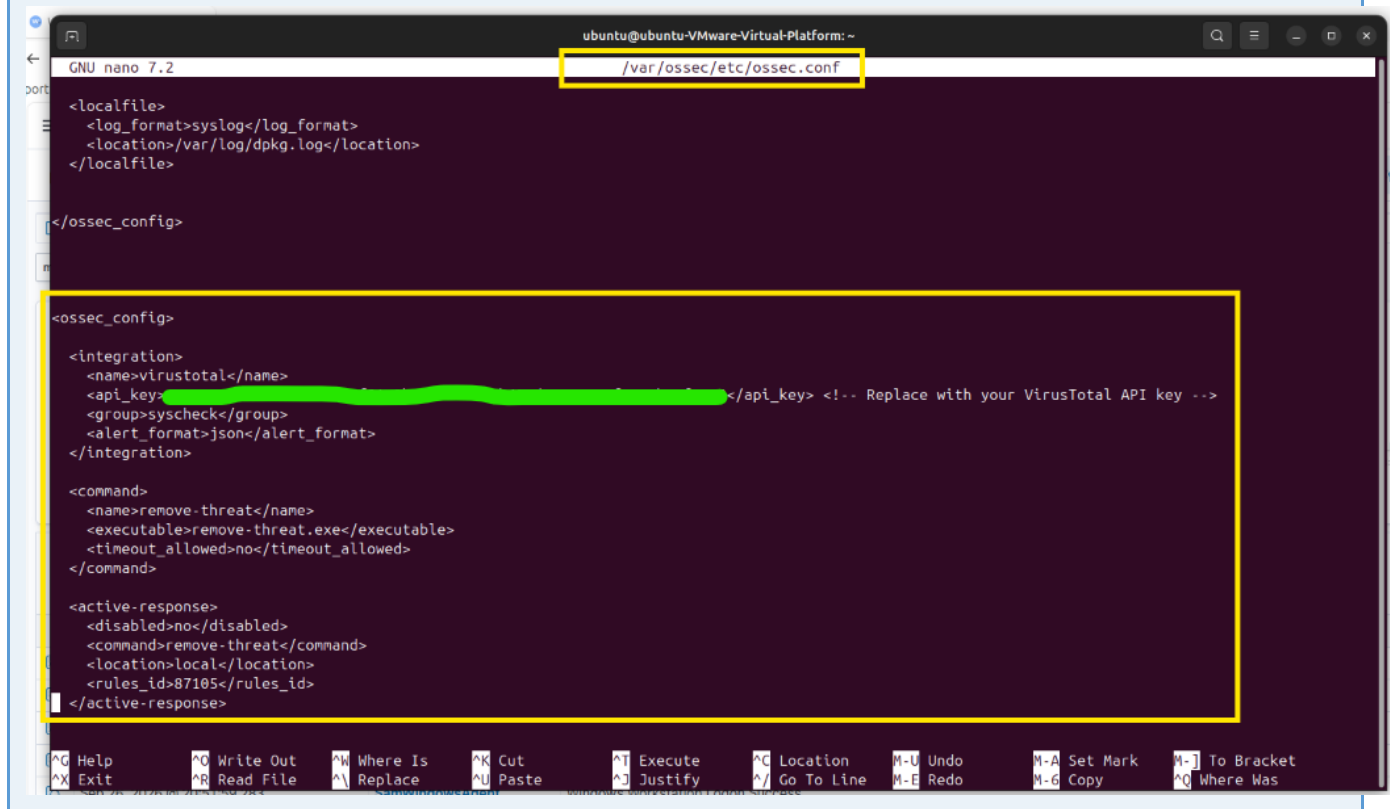
<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
```

```

<location>local</location>
<rules_id>87105</rules_id>
</active-response>

```

FIGURE 2: Manager configuration



Common issue

The command name and executable name must match exactly. location=local runs the executable on the agent that generated the alert.

5. Create and Convert the Removal Script

Wazuh sends the alert to the executable as JSON through standard input. The script extracts the VirusTotal source path, removes the file, and records the result.

```

import sys
import json
import os
from datetime import datetime

LOG_FILE = r"C:\Program Files (x86)\ossec-agent\active-response\active-responses.log"

def log(message):
    with open(LOG_FILE, "a", encoding="utf-8") as file:
        file.write(f"{datetime.now():%Y/%m/%d %H:%M:%S} remove-threat.exe: {message}\n")

try:
    log("Started")
    data = json.loads(sys.stdin.readline())

    if data.get("command") != "add":
        log("Invalid command")
        sys.exit(0)

```

```

file_path = data["parameters"]["alert"]["data"]["virustotal"]["source"]["file"]

if os.path.isfile(file_path):
    os.remove(file_path)
    log(json.dumps(data) + " Successfully removed threat")
else:
    log(json.dumps(data) + " Error removing threat: file not found")

except Exception as error:
    log(f"Error removing threat: {error}")
    sys.exit(1)

python -m PyInstaller -F .\remove-threat.py

```

FIGURE 3: PyInstaller build

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Demo Windows> cd .\Desktop\
PS C:\Users\Demo Windows\Desktop> python -m PyInstaller -F .\remove-threat.py
275 INFO: PyInstaller: 6.22.3, contrib hooks: 2026.7
276 INFO: Python: 3.14.7
306 INFO: Platform: Windows-10-10.0.19045-SP0
306 INFO: Python environment: C:\Users\Demo Windows\AppData\Local\Python\pythoncore-3.14-64
309 INFO: wrote C:\Users\Demo Windows\Desktop\remove-threat.spec
315 INFO: Module search paths (PYTHONPATH):
['C:\\Users\\Demo Windows\\Desktop',
 'C:\\Users\\Demo '
 'Windows\\AppData\\Local\\Python\\pythoncore-3.14-64\\python314.zip',
 'C:\\Users\\Demo Windows\\AppData\\Local\\Python\\pythoncore-3.14-64\\DLLs',
 'C:\\Users\\Demo Windows\\AppData\\Local\\Python\\pythoncore-3.14-64\\Lib',
 'C:\\Users\\Demo Windows\\AppData\\Local\\Python\\pythoncore-3.14-64',
 'C:\\Users\\Demo '
 'Windows\\AppData\\Local\\Python\\pythoncore-3.14-64\\Lib\\site-packages',
 'C:\\Users\\Demo Windows\\Desktop']
795 INFO: checking Analysis
930 INFO: checking PYZ
961 INFO: checking PKG
966 INFO: Bootloader C:\Users\Demo Windows\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\PyInstaller\bootloa
der\Windows-64bit-intel\run.exe
966 INFO: checking EXE
977 INFO: Rebuilding EXE-00.toc because remove-threat.exe missing
978 INFO: Building EXE from EXE-00.toc
979 INFO: Copying bootloader EXE to C:\Users\Demo Windows\Desktop\dist\remove-threat.exe
1000 INFO: Copying icon to EXE
1017 INFO: Copying 0 resources to EXE
1019 INFO: Embedding manifest in EXE
1028 INFO: Appending PKG archive to EXE
1109 INFO: Fixing EXE headers
1246 INFO: Building EXE from EXE-00.toc completed successfully.
1248 INFO: Build complete! The results are available in: C:\Users\Demo Windows\Desktop\dist
PS C:\Users\Demo Windows\Desktop>

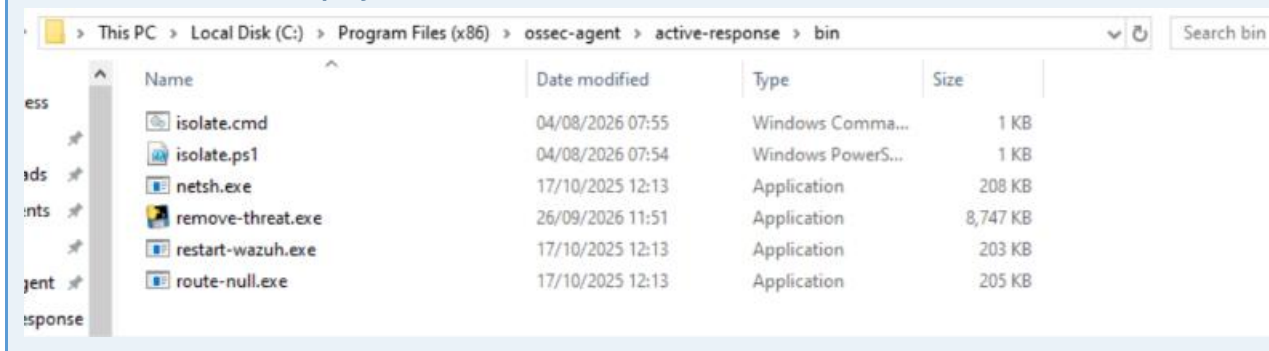
```

The resulting file was copied to:

In the **dist folder** there will **remove-threat.exe** file, cut and paste that file in following location

```
C:\Program Files (x86)\ossec-agent\active-response\bin\remove-threat.exe
```

FIGURE 4: Executable deployment



Before running test or downloading the EICAR test file make sure to restart both the *Wazuh agent* and *Wazuh Manager*.

6. Test and Results

The EICAR test file was downloaded into the monitored Downloads directory. Wazuh produced the expected detection and response sequence.

FIGURE 5: Threat Hunting evidence

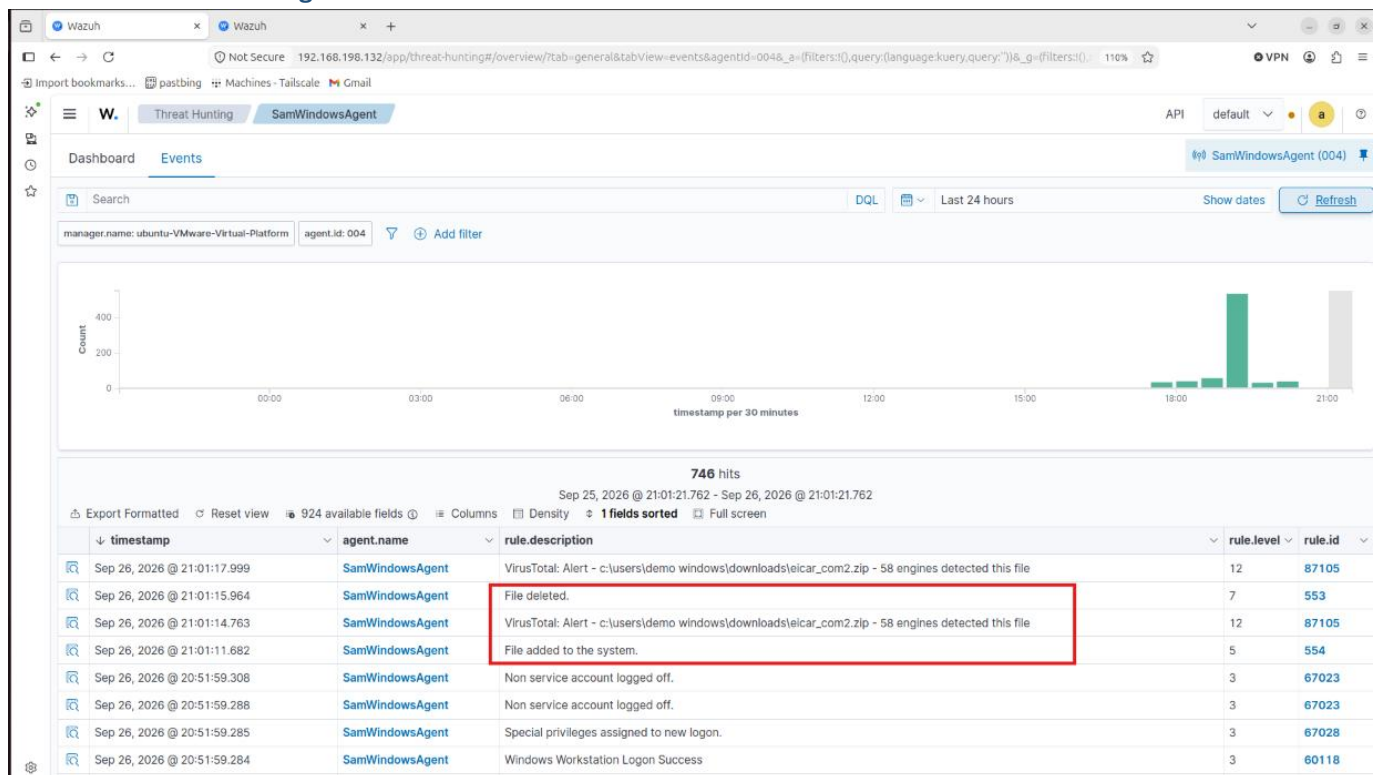


FIGURE 6: Successful malicious file deleted logs from windows agent.

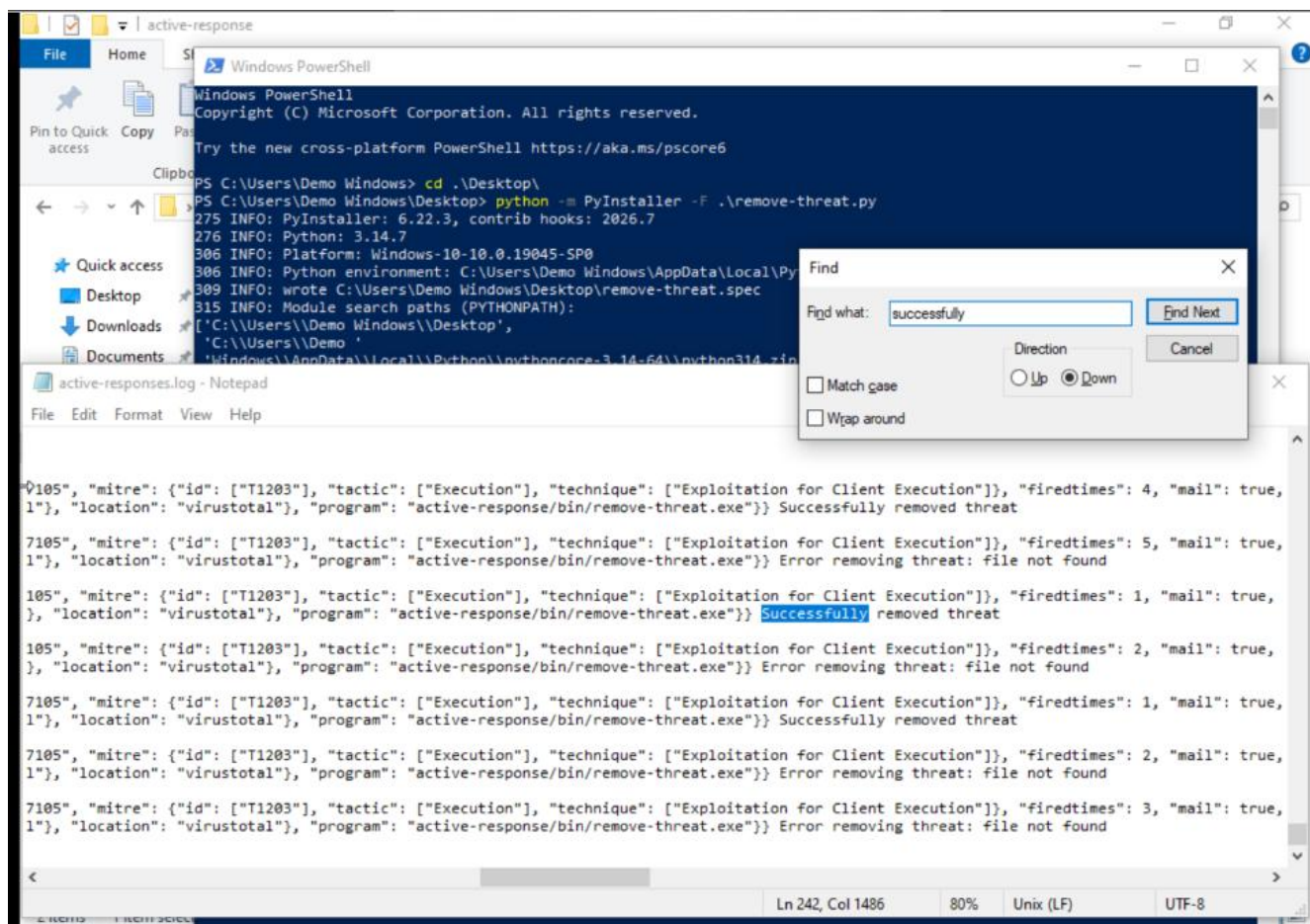


Figure 8. Windows active-responses.log confirms “Successfully removed threat”; later attempts report “file not found”.

IMPORTANT: Why alerts continue after deletion

VirusTotal alerts can appear after rule 553 because Wazuh stores alerts as historical evidence and multiple FIM events may already have submitted the same file hash. In this test, file creation and checksum change produced separate syscheck events. The first Active Response deleted the file successfully. Later rule 87105 events triggered the executable again, so it correctly reported “file not found”. This does not mean that the file returned or that the first deletion failed.

7. Understanding the Rule IDs and Levels

A rule ID identifies the event type. A rule level shows the event’s importance. A higher level generally requires more attention, but the level does not indicate whether deletion succeeded or failed.

Rule ID	Level	Meaning	Use this lab
550	7	Integrity checksum changed	File content or hash changed.
553	7	File deleted	FIM observed that the file was removed.
554	5	File added	FIM observed a new file.
657	Varies	Active Response log event	Parent events are required by custom rules.
87105	12	VirusTotal malicious-file alert	Triggered remove-threat.exe.
100092	12	Custom removal success	Optional dashboard confirmation; not observed.
100093	12	Custom removal error	Can appear on duplicate

			attempts after deletion.
--	--	--	--------------------------

Level 7 vs Level 12

Level 7 represents a potentially security-relevant event that needs context. Level 12 is a high-importance event that may indicate an attack. Rule 553 being level 7 does not mean deletion failed; it only classifies the deletion event severity.

8. Troubleshooting

- No rule 554: verify the monitored path, realtime setting, agent connection, and agent restart.
- No rule 87105: verify internet, DNS, API key, rate limiting, and manager integration logs.
- 87105 appears but the file remains: verify location=local, the EXE name/path, and active-responses.log.
- “File not found” after success: duplicate 87105 events retriggered the executable after the first successful deletion.
- No 100092: confirm collection and decoding of active-responses.log through rule 657. This alert is optional evidence.

9. Conclusion

The practical was successful. Wazuh detected the EICAR file, VirusTotal classified the hash as malicious, the Windows Active Response removed the file, rule 553 recorded deletion, and active-responses.log recorded “Successfully removed threat”. Duplicate VirusTotal alerts and later “file not found” messages were expected consequences of multiple queued FIM events, not evidence of a failed response.

Final status

DETECTION: Successful | ACTIVE RESPONSE: Successful | FILE DELETION: Confirmed

References

- Wazuh documentation: Detecting and removing malware using VirusTotal integration.
- Wazuh documentation: Custom Active Response scripts.
- Wazuh documentation: Rules classification.
- EICAR: Anti-malware test file.